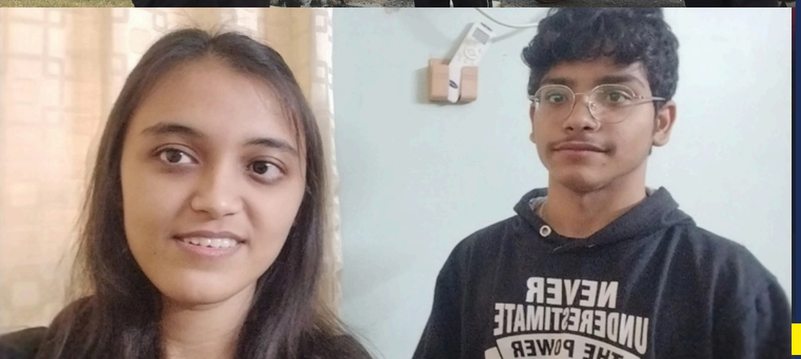
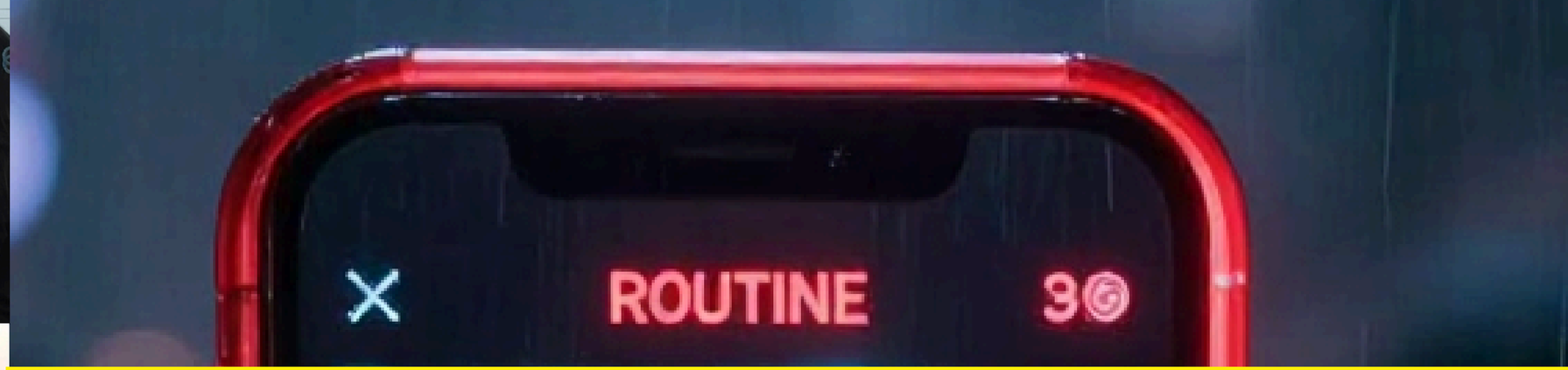










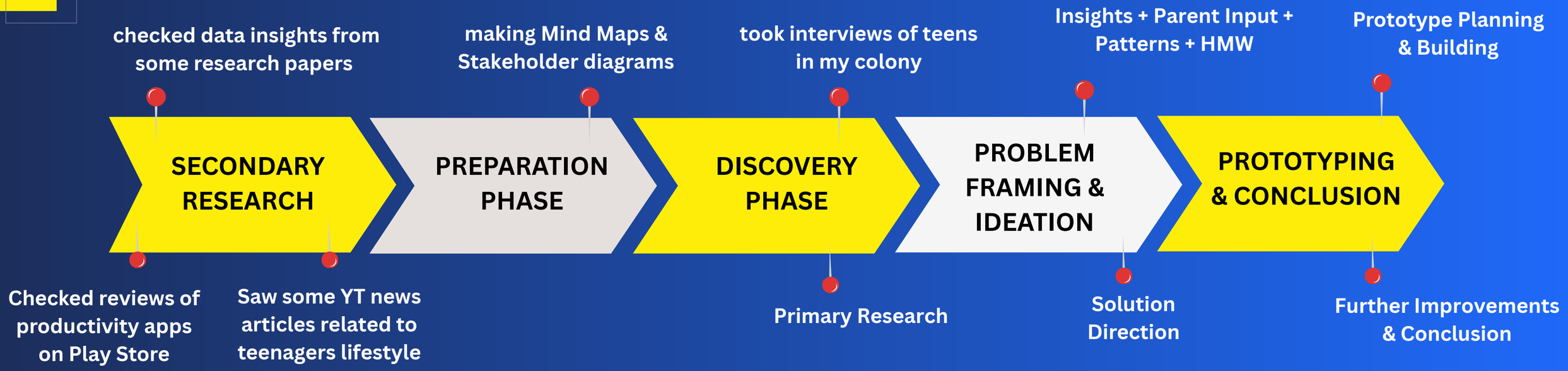
HABIT TRACKER APP FOR TEENS

• SRIJITA CHAKRABORTY

Problem Statement: (5) People want to build healthy habits but struggle to sustain them within the constraints of their daily routines.

- *Focused on a particular age group (teens: 13-19). Understanding the challenges they face due to academic pressure, distractions and declining attention span.*

Research Plan:



- I decided to begin by selecting the problem statement around teen habit sustainabilty as I everyday observe the routine and discipline challenges of my sister (17 yrs) and kids around her age during exam preparation phases. Since the topic was broad and I had limited time, I first decided to understand the current ecosystem of productivity & habit tool apps on Play Store as well as behavioral trends affecting teenagers.

To structure my understanding and avoid assumption bias, I outlined the research stages and identified key stakeholders influencing teen routines such as parents, teachers, schools, and digital platforms.

- I then conducted secondary research including app market scans and behavioral studies to understand existing solutions and documented gaps. After this landscape understanding, I proceeded with primary research through teen interviews and pilot discussions to identify real behaviour patterns and narrow the research focus for deeper analysis.

Secondary Research:

- I conducted secondary research to understand teen habit behaviour, attention challenges, and routine sustainability patterns. This included reviewing productivity and habit tracking apps available on the Play Store as well as reading published research papers related to adolescent behavior, habit formation, attention span, and technology impact.

- The goal of secondary research was to understand the current solution landscape and validate whether routine inconsistency and attention fragmentation among teenagers is supported by existing research.

What Habit Sustainability Means in This Context?

- In this study, habit sustainability refers to a teenager's ability to start and maintain consistent daily routines such as study schedules, sleep timing, focus practices, and healthy lifestyle habits over time.

It includes:

- *consistency in study routines*
- *focus and attention control*
- *screen-time discipline*
- *self-regulation behavior*
- *recovery after routine breaks*

Why Habit Sustainability Is Important for Teens?

Secondary literature shows that consistent habit behavior in adolescence supports:

- *improved academic performance*
- *stronger self-regulation*
- *better stress handling*
- *improved attention control*
- *long-term discipline development*

Teen years are a key behavior-formation stage.

Key Insights from Research Papers Reviewed:

- ***Adolescent Health Behaviors and Difficulty Concentrating, Remembering, and Making Decisions:***
  This research shows that many adolescents experience problems with concentration, memory, and decision-making, factors closely tied to attention issues and routine struggles.

- ***How to Form Good Habits: This longitudinal research tracks habit formation over time and shows how consistent behavior leads to stronger habits useful when discussing why routines break or stick.***

- ***Digital Overuse & Cognitive Decline (Brain Rot): This study directly links heavy digital usage (like excessive social media) to reduced attention span, decreased cognitive control, and automatic habitual screen checking.***

Impact of ICT Exposure on Executive Functions in Youth

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A1 Author (Year)

	A	B	C	D	E	F	G	H	I	J
1	Author (Year)	Study Design	Country	Sample Size	Age Range/Mea	ICT Type	Assessed Execu	Key Findings	Parental/Family	Source
2	Mallawaarachchi	Systematic revie	Australia	176742	Early childhood (	Television, video	Working memory	TV exposure ass	Co-use with othe	1
3	Massaroni et al.	Systematic revie	Italy	32274	0 to 7 years	Televisions, com	Working memory	Preschool scree	Watching televisi	1
4	Song et al. (202	Longitudinal coh	USA	11815	9–10 years	Video entertainm	Inhibitory control	Excessive time c	Public health stra	1
5	Bustamante et a	Meta-analysis	Spain	6922	0 to 6 years	TVs, computers/	Working memory	No statistically si	Aims to estimate	1
6	Kim and Tsethlik	Longitudinal coh	South Korea	2150	5.59 months	TVs, tablets, sm	Plan/organize, In	Higher levels of	Positive parentin	1
7	Bukhalenkova et	Longitudinal coh	Russia	490	5–7 years old	TVs, smartphone	Working memory	Children who wa	53.7% watched	1
8	Law et al. (2023)	Longitudinal coh	Singapore	437	8.84 years	Mobile electronic	Naming inhibitior	Screen use may	Screen time serv	1
9	Soltani Kouhban	Quasi-experimer	Iran	133	8–12 years old	TVs, laptops/con	Brain wave funct	Function of brain	Evaluated the H	1
10	Choi et al. (2024	Longitudinal coh	Canada	110	9.14 years	TVs, tablets, con	Learning, Execu	Shorter sleep du	Positive parentin	1
11	Rai et al. (2023)	Quasi-experimer	Canada	44	3.5 years	YouTube (TV) ar	Working memory	Excessive scree	Children spent 4	1

Adolescent Health Behaviors and Mental Health Outcomes

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J5

	A	B	C	D	E	F	G	H
1	Health Behavior	Variable Type	Measurement Ur	Odds Ratio (Adj	95% Confidence	Statistical Signifi	Association with	Source
2	Physical activity	Binary	60 minutes per d	0.79	0.67-0.95	0.013	Lower odds	1
3	Breakfast consu	Binary	Consume everyc	0.71	0.63-0.80	< .001	Lower odds	1
4	Smoking	Binary	Anytime in past 3	1.40	1.15-1.70	< .001	Higher odds	1
5	Alcohol consump	Binary	Anytime in past 3	1.29	1.17-1.43	< .001	Higher odds	1
6	Sleep	Binary	8 hours per day	0.75	0.64-0.87	< .001	Lower odds	1
7	Multiple health b	Continuous (per	Sum of positive b	0.73	0.68-0.78	< .001	Lower odds	1
8	Vegetable consu	Binary	At least 3 times p	1.44	0.98-2.13	Not in source	No significant as	1

(Data insights collected from research papers) links are provided in the last slide (References)

Key Insights from Secondary Research: Common patterns observed

- attention span decline among youth
- digital distraction reduces sustained focus
- habit success depends on environment + structure
- motivation alone is insufficient for habit continuity
- self-regulation skills are critical
- technology overuse impacts routine stability

Gap Identified:

App market scan + literature review showed:
most habit apps are

- adult-focused
- tools assume high discipline
- interfaces are complex for teens
- teen-specific habit behavior is not directly addressed

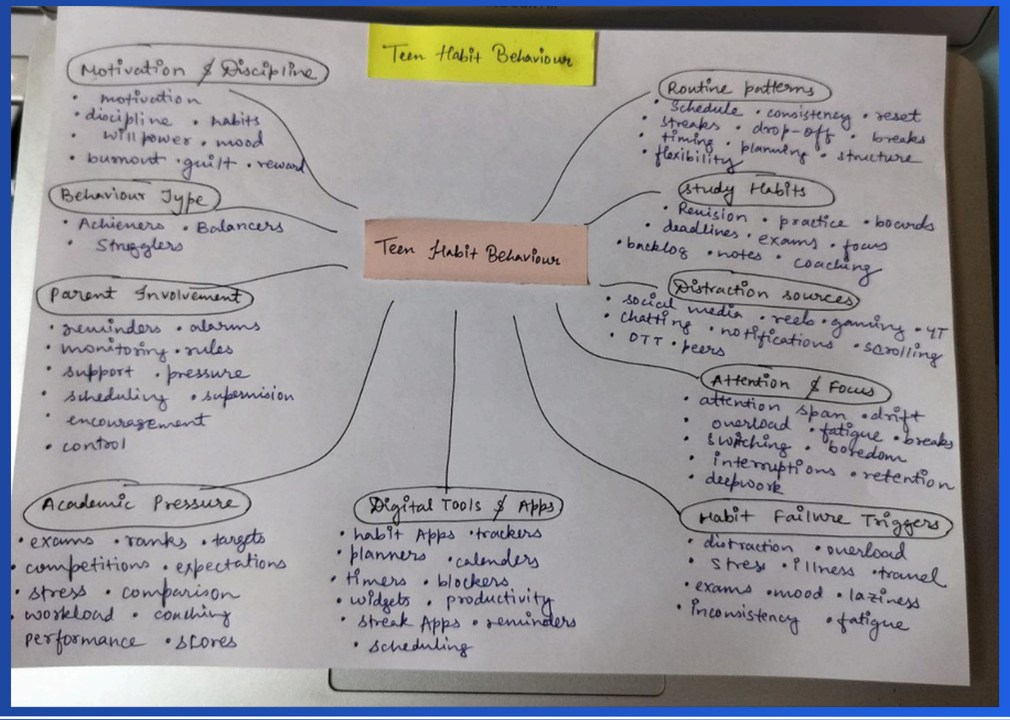
- ***This justified the need for a teen-focused, minimalist habit tracker approach.***

Preparation Phase:

- Once I developed an initial understanding of the topic through secondary research and app landscape review, I prepared for deeper user research by structuring my assumptions and knowledge about teen habit behavior and routine sustainability.
- To organize my thinking and avoid bias, I created a mind map and a stakeholder map to clearly outline influencing factors, behavior drivers, and possible intervention points in teen routine formation.

Mind Map

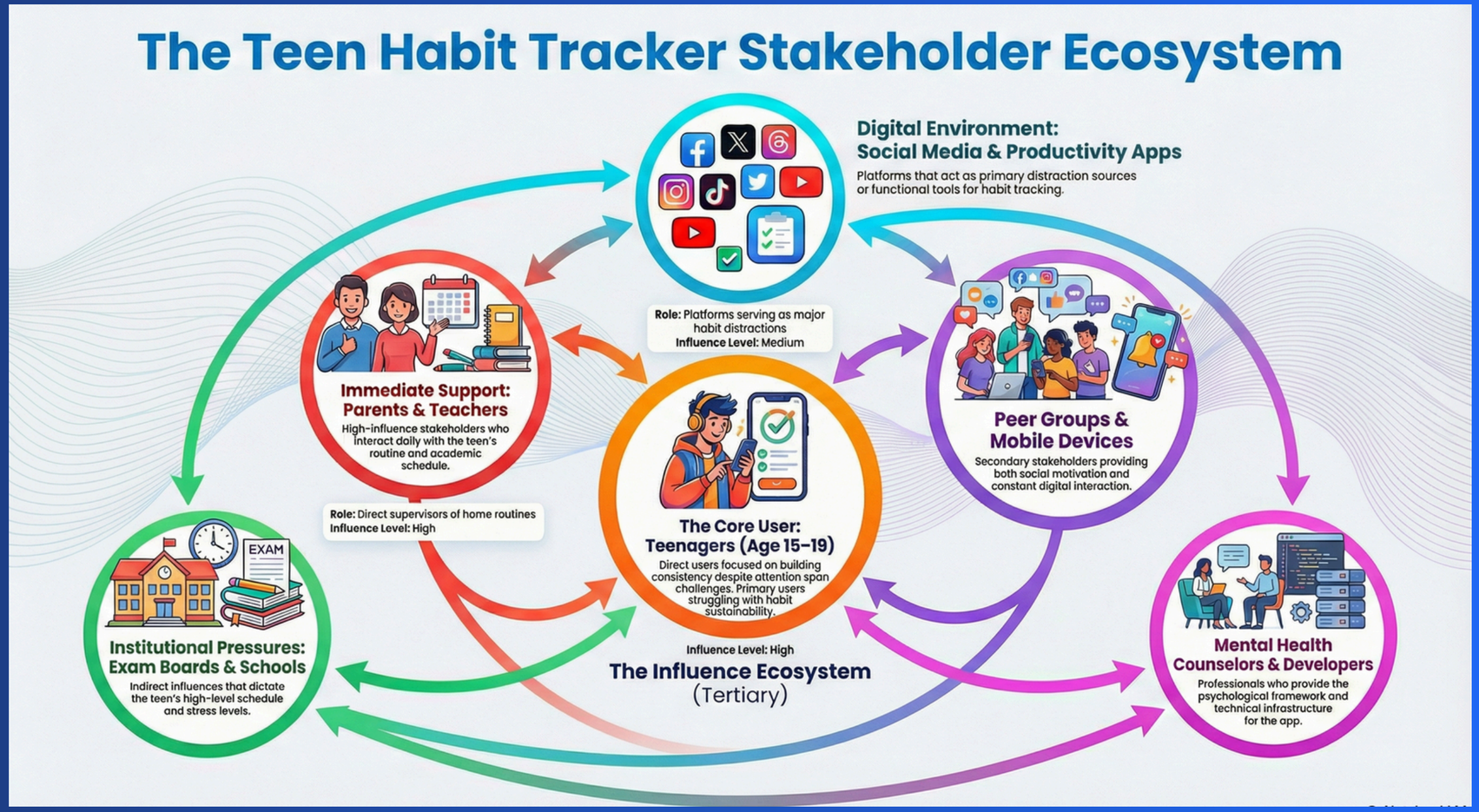
- Based on secondary research and initial observations, a mind map was created to articulate key dimensions of teen habit behaviour.



Stakeholder Diagram

- After the mind map, a stakeholder diagram was created to identify all major stakeholders interacting with the teen user's routine ecosystem. I jotted down all the stakeholders would be involved in this and gave prompts in Notebooklm for making the Stakeholder diagram.

- ***This helped understand external influences on habit behaviour.***



Pilot Study

- Since the topic of habit sustainability is broad, I conducted a small pilot study to narrow the research focus and identify recurring behavior patterns.
- I interviewed a small group of teenagers in my township before conducting deeper structured analysis (14-19)

Pilot Study Participants

- Teenagers from Classes 6–12
- Age group: approximately 14–19
- Mixed academic performance levels
- Mixed discipline styles
- Video interviews recorded
- Responses anonymized for ethical reasons

Primary research proof includes recorded video responses.

Pilot Study Research Questions

Participants were asked open-ended behavior questions such as:

- How do you usually plan your day?
- Do you follow a daily routine?

- What habits have you tried to build?
- Where do routines usually break?
- What distracts you most?
- Do you use any habit or productivity apps?
- If a teen-specific habit app existed, how would you use it?
- What makes you stop following routines?
- How do exams affect your habits?
- Who helps manage your routine - you or parents?

Parent Stakeholder Input (Discovery Extension)

Since many teens depend on parents for routine regulation, an additional stakeholder discussion was conducted with parents to understand:

- parent role in teen routine control
- alarm & schedule support behavior
- monitoring habits
- discipline enforcement patterns
- external motivation triggers
- This added stakeholder-side behavioral insight.

I asked my mother and some other aunts in my colony.

Discovery Phase (Primary Research):

Sampling

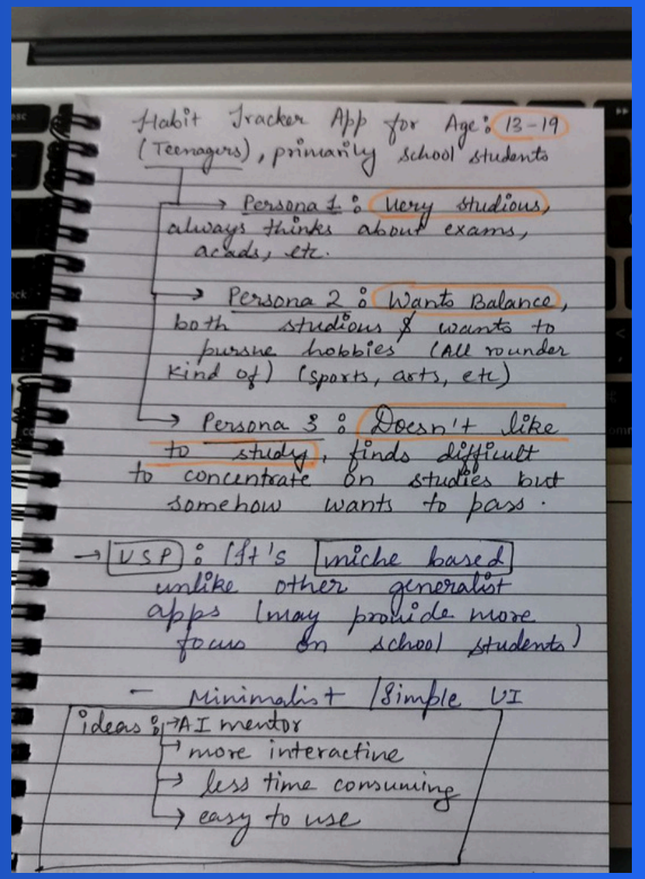
For the discovery phase, I conducted primary research focused on teenagers and young students to understand their daily routines, habit patterns, distractions, and consistency challenges.

Participants interviewed:

- 8-10 teenagers (school students)
- Age group: 14-19 years
- Mixed academic backgrounds
- Included both high-performing and average-performing students
- Included students preparing for board exams and entrance exams

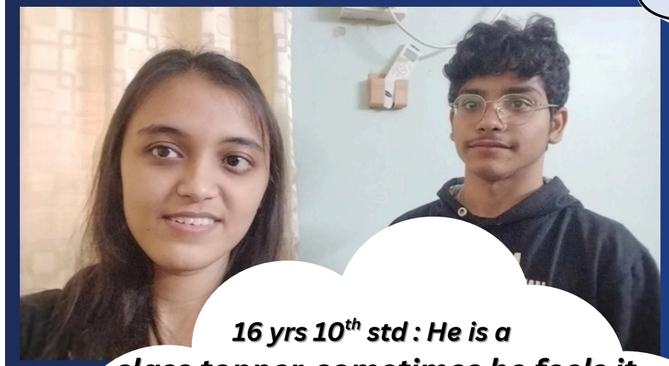
This helped me capture differences between:

- Highly structured students
- Irregular routine students
- Digitally distracted students



The Stories That Made Me Realise : Teenagers Struggle with Consistent Routines

Routine Consistency



16 yrs 10th std : He is a class topper, sometimes he feels it very burdening to keep up with the expectations which leads to hectic daily routine, finds social media distracting



14 yrs old 8th std : They literally don't find study interesting, specially to sit at a place, that's why they hate 4 PM - 7PM as their parents tell them to study that time



18-19 yrs old 12th std : They just somehow wants to pass their entrance exams to get a college, finds pressure among peers as they find 12th std is all about competitions and exams, always ending up thinking they'll start from tomorrow

Digital Distraction



17 yrs old 11th std : They want balance between studies and their hobbies, whenever they do some extracurricular activity, got scolded by their guardians, which they find very overwhelming, tend to miss even after putting alarms

Primary Methods Involved:

Contextual Inquiry

I interacted with students in their natural environment (home / study setting / daily routine context) instead of a formal survey-only approach.

This helped me understand:

- How they actually plan their day
- How routines break in real life
- How they react when habits fail
- Their real interaction with phones and apps

Observation

I compared: What users say vs what they actually do

I observed:

- Study vs distraction time
- Screen usage patterns
- Sleep timing irregularity
- Habit planning vs execution gap
- Dependency on reminders from parents

Interview Focus Areas

I asked questions around:

- Daily routine structure
- Study habits
- Sleep schedule
- Screen time usage
- Distraction triggers
- Existing habit tracking methods
- Motivation drop patterns
- Use of productivity apps
- Parental involvement in routine
- Why habit tracking fails?

Research Objectives:

- To understand routine instability among teenagers
- To understand habit formation challenges
- To study attention loss due to digital overuse
- To identify why existing habit tracker apps are not used consistently
- To understand motivation patterns in teens
- To explore need for teen-specific habit tracker design
- To understand reminder and accountability needs

User Personas

From the research, we derived 3 key personas:

- Achievers: Exam-Focused Structured Teen
- Balancers: Distracted Multi-Tasker Teen
- Strugglers: Motivation-Dependent Teen

Each persona shows:

- Different routine stability
- Different distraction levels
- Different motivation triggers

User Journey Maps (Routine Mapping)

To understand daily life patterns, we mapped routine journeys.
We observed two major day types:

Study-Heavy Days

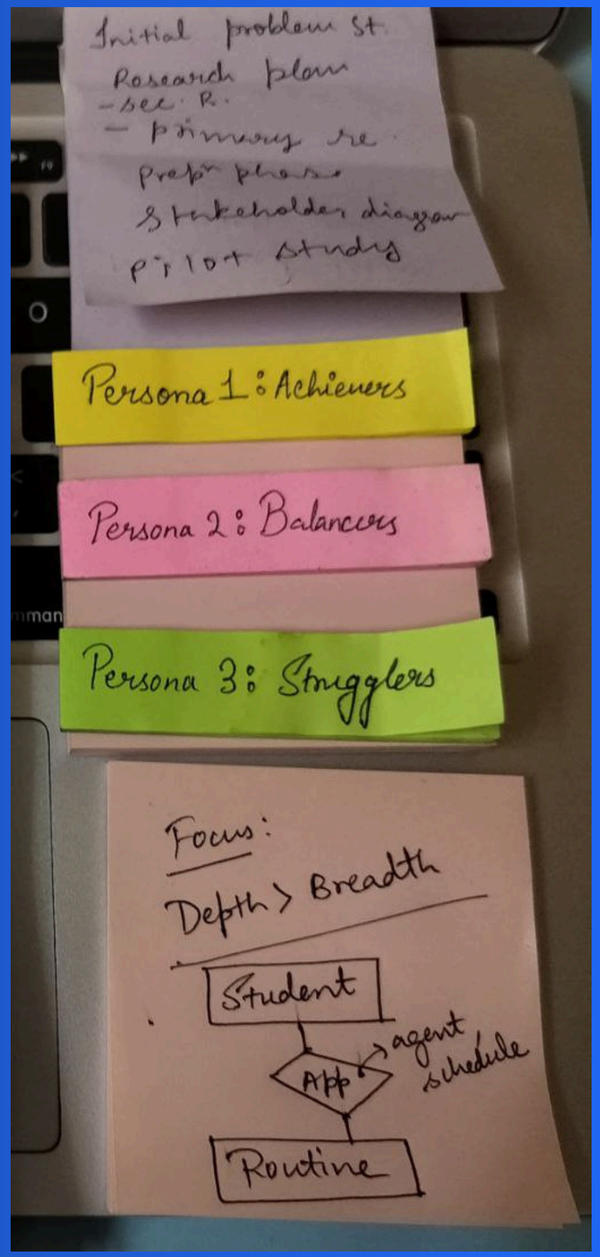
- **Tight schedule**
- **High stress**
- **Habit breaks**
- **Sleep sacrifice**

Unstructured Days

- **Flexible time**
- **High screen usage**
- **Low habit follow-through**
- **Poor time allocation**

Key Findings for Journey Mapping

- Sleep schedules are highly irregular
- Habits are started but rarely sustained
- Screen usage is constant throughout the day
- Teens rely on memory instead of systems
- Motivation drops after 3–5 days of habit attempts
- Free time is often wasted unintentionally



Empathy Mapping

Users	Needs	Insights
Persona 1 : Achievers	• Minimal distraction environment • Fast, low-friction habit logging • Academic goal alignment • Progress tracking without gamification overload • Custom study-routine blocks • Offline / low-app dependency options	• Already disciplined, don't want "extra app burden" • Prefer simple trackers over feature-heavy apps • Use phone mainly for academic content • Habit tools must be lightweight or they won't adopt • Value consistency > motivation • Risk: reject habit apps if they feel childish or noisy
Persona 2 : Balancers	• Study + hobby routine balance • Visual schedule planning • Flexible habit tracking • Reminder nudges • Streak & progress motivation • Time-block suggestions	• Want structure but not rigidity • Often over-plan and under-execute • Motivated by streaks & visual progress • Respond well to soft reminders • Switch between productivity & leisure frequently • Need adaptive routine support
Persona 3 : Strugglers	• Strong reminders • Micro-habit breakdown • Motivation triggers • Distraction control support • Simple daily checklist • Parent/mentor accountability option	• Start habits but drop quickly • High screen distraction • Low routine consistency • Feel guilt after missed days • Need guided structure • Motivation driven, not discipline driven • Require very small starting steps

Insights - Routine & Habit Formation Study

After completing our secondary research and primary interviews with teenagers, I summarized the following key behavioural & routine insights:

- Almost all teenagers rely heavily on mobile phones throughout the day for both academic and non-academic activities
- High screen time is directly affecting attention span and routine consistency
- Many students plan routines but fail to execute them consistently
- Habit building is usually motivation-driven, not system-driven
- Students start habits with enthusiasm but drop them within a few days
- Most teenagers do not use dedicated habit tracker apps, they rely on memory, parents, or school pressure
- High-performing students (achievers) prefer simple, distraction-free systems instead of feature-heavy apps
- Balanced students want to manage studies + hobbies, but struggle with time allocation
- Struggling students show low routine discipline and high distraction patterns
- Sleep schedules are highly irregular, especially during exams and assignment periods
- Many teens depend on parents for waking up and routine enforcement
- Students feel guilty after wasting time but repeat the same distraction patterns
- Reward systems, streaks, and visual progress tracking increase motivation temporarily
- Overly complex productivity apps lead to quick abandonment
- Teenagers respond better to small habit steps and simple daily checklists than big long-term plans

These insights indicate that teenagers need a simple, low-friction, teen-focused habit tracking system that supports small, consistent actions instead of complex productivity structures.

HMW Statements - Teen Habit Tracker Study

Based on my research insights, I framed the following How Might We (HMW) statements to translate user challenges into actionable design opportunities:

- How might we help teenagers build small, consistent daily habits so that they can sustain routines without feeling overwhelmed?
- How might we design a teen-focused habit tracker that is simple and low-friction so that users do not abandon it after a few days?
- How might we reduce digital distractions for teenagers while still using the mobile phone as a habit-building tool?
- How might we support students who start habits with motivation but fail to maintain consistency over time?
- How might we create a habit system that works for different teen types - achievers, balancers and strugglers instead of a one-size-fits-all model?
- How might we help teenagers visually track progress and streaks so that they feel encouraged to continue their routines?
- How might we design reminder and nudging systems that motivate teens without becoming annoying or ignorable?
- How might we support teens with irregular sleep and study schedules to gradually stabilize their daily routine?
- How might we involve light accountability (parents/mentors/self-check) so that struggling teens stay on track?
- How might we convert large goals like “study regularly” into micro-habits that are easier to start and maintain?

Brainstorming & Ideation

After framing the HMW statements,
Through brainstorming, I generated multiple idea clusters and feature concepts.

Idea Cluster 1 - Simple Teen Habit Tracker
Focus: Make habit tracking low-friction and teen-friendly
Possible Components:

- One-tap daily habit check-in
- Micro-habit creation (very small steps)
- Visual streak tracker
- Minimal UI to avoid overload
- Quick daily checklist view

Idea Cluster 2 - Motivation & Consistency Support

- *Focus:* Help teens maintain habits beyond initial motivation
- *Possible Components:*
- Streak rewards & progress badges
- Gentle reminder nudges
- Missed-day recovery system
- Weekly progress feedback
- Visual growth charts

Idea Cluster 3 - Distraction & Routine Control
Focus: Reduce digital distraction impact on routines
Possible Components:

- Focus mode during study blocks
- Screen-time awareness prompts
- Habit-before-entertainment rule option
- Study-time timers
- Distraction alerts

Idea Cluster 4 - Persona-Based Routine Modes
Focus: Support different teen types (Achievers, Balancers, Strugglers)
Possible Components:

- Mode selection during onboarding
- Different reminder intensity levels
- Flexible vs strict tracking styles
- Hobby + study balance templates
- Parent-assisted mode (optional)

Down Selection Process - Final Design Direction

To select the most suitable final solution direction, I applied a structured down-selection method using the following evaluation criteria:

Selection Criteria

C1 — **Desirability**
Does this solution address a real and frequent teen routine problem?

C2 — **Feasibility**
Can this be built within available time and technical scope?

C3 — **Viability**
Can this realistically be adopted and used regularly by teenagers?

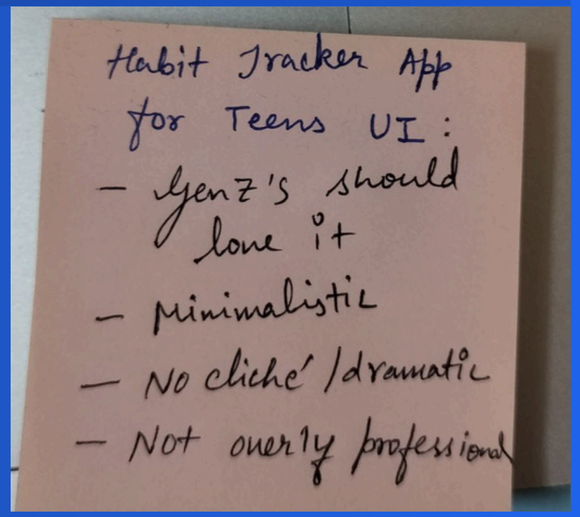
C4 — **USP (Uniqueness)**
Is this specifically designed for teens instead of generic productivity users?

C5 — **Flexibility**
Can it adapt to different routine styles and persona types?

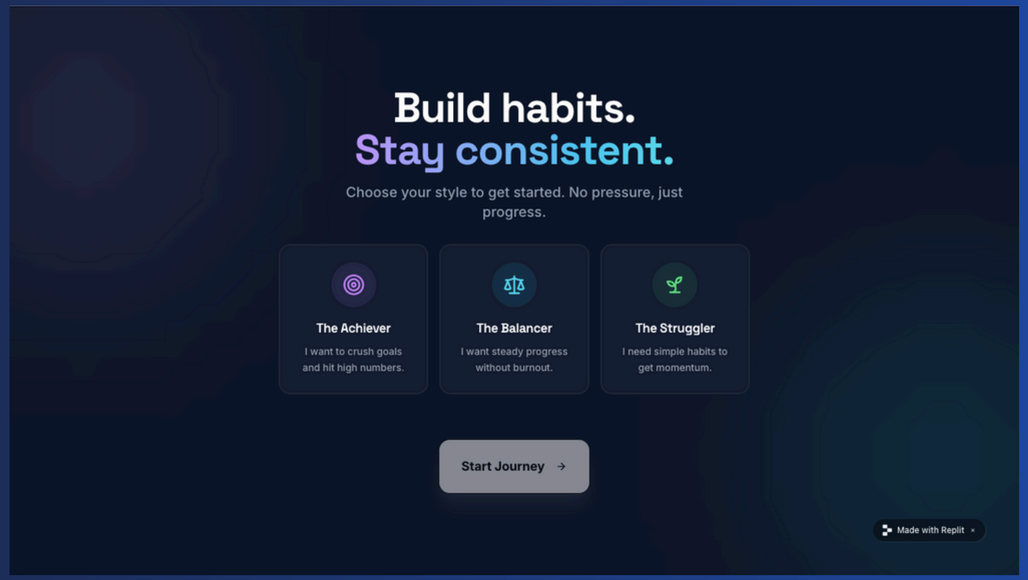
C6 — **Accessibility**
Is it simple and usable for teens with different academic levels and tech comfort?

C7 — **Ethics & Privacy**
Does it protect teen data and avoid over-surveillance or pressure?

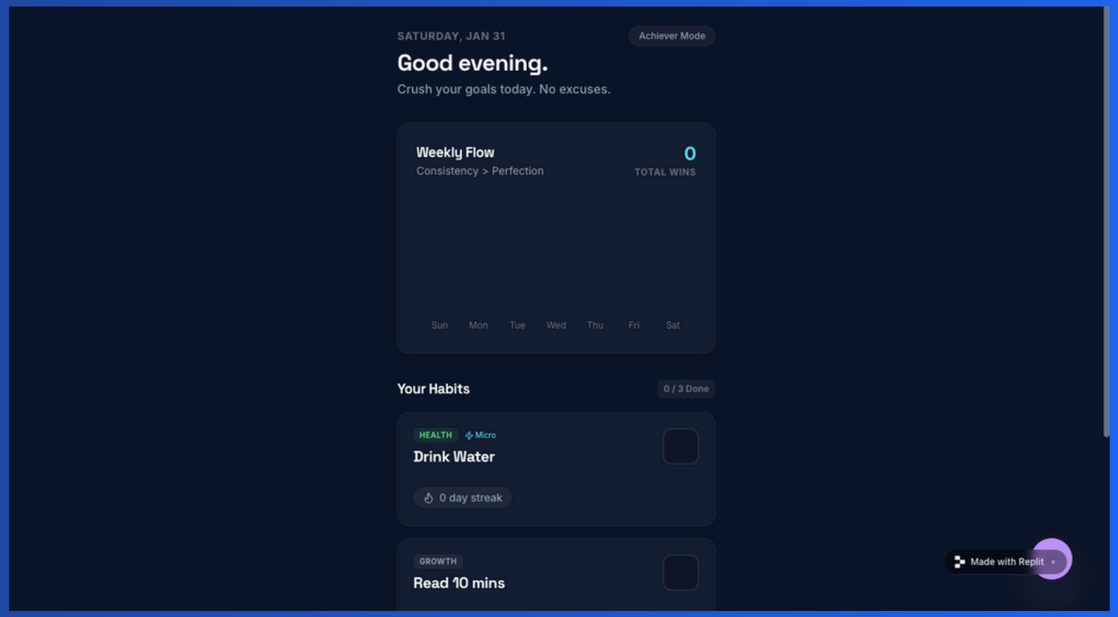
Down Selection Outcome: *Based on the evaluation, I selected the direction of a teen-focused, low-friction habit tracker app with persona-based modes and micro habit support as the final design solution.*



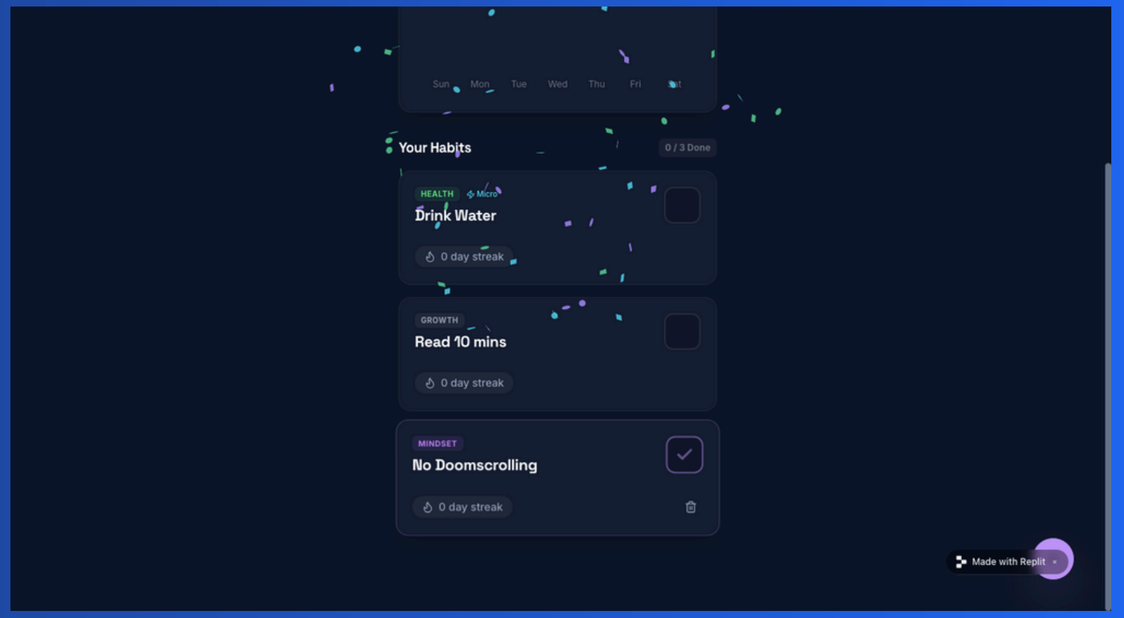
Prototyping



persona-based modes



Micro-habit support



Do checkout (paste the below link on your browser)

https://habit-spark--srijitachakrab3.replit.app

Future Scope & Limitations

- Study focused on depth of behavioral research over full product and business validation due to time constraints
- Primary research sample size was limited and short-term
- No long-duration habit retention tracking was conducted
- Prototype was not tested through large-scale real user trials
- Revenue model and monetization strategy were kept out of scope, would be implemented further along with market analysis
- Technical scalability and advanced system design were not explored
- Future work can include larger teen user studies and longitudinal tracking
- Pilot testing with real teenagers to measure habit consistency impact
- AI-based personalization and smarter nudging systems can be added
- Market positioning and school/coaching partnerships can be explored

Conclusions & Learnings

- Teen habit failure is driven more by distraction and routine friction than lack of intention
- Different personas (Achievers, Balancers, Strugglers) need different support styles
- One-size-fits-all productivity tools are less effective for teenagers
- Low-friction tracking increases adoption likelihood
- Micro-habits reduce starting resistance
- Visual streaks and progress tracking improve short-term motivation
- Research-first approach leads to clearer and more grounded solution design
- Depth-focused user understanding is more valuable than feature-heavy design
- Persona-driven product thinking improves relevance of solutions

Bloopers !







They were telling “Kucch balance nahi hota”

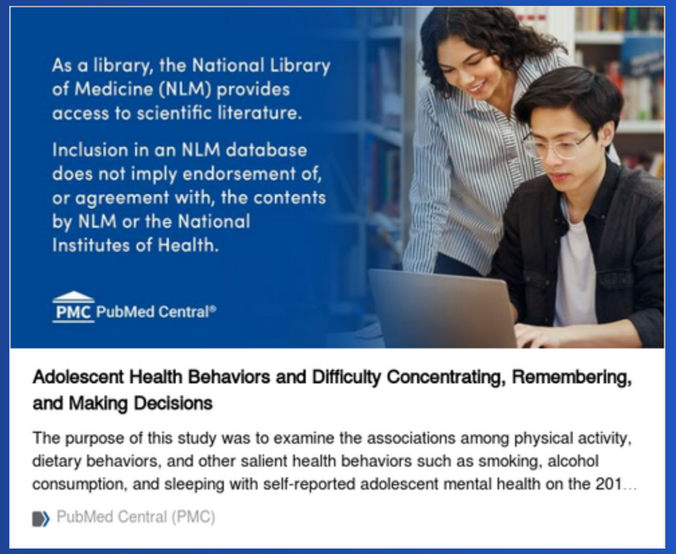


These kids were really scared of me! Hopefully I’ll work further on child safety on some other case study if till that time I don’t end up becoming a Minister!

References

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IMPACT OF SCREEN TIME ON ATTENTION SPANS: EXPLORING DURATION, CONTENT, AND CONTEXT ACROSS VARIOUS AGE...

Background: In the digital era, screen time has become ubiquitous in daily life, raising concerns about its potential impact on attention spans across various age groups. Existing research presents mixed findings, with some studies indicating a...

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